(i)

$$X * Y = \lambda \mathbf{x} + (1-\lambda)\mathbf{y}$$

$$Y * X = \lambda \mathbf{y} + (1-\lambda)\mathbf{x}$$

$$\begin{aligned} X * Y = Y * X &\implies \lambda \mathbf{x} + (1-\lambda)\mathbf{y} = \lambda \mathbf{y} + (1-\lambda)\mathbf{x} \\ &\implies (2\lambda - 1)(\mathbf{x} - \mathbf{y}) = \mathbf{0} \end{aligned}$$

$\because X, Y$ are distinct $\therefore \mathbf{x} \neq \mathbf{y}$

$$2\lambda - 1 = 0 \implies \lambda = \frac{1}{2}$$

$X * Y$ and $Y * X$ are distinct unless $\lambda = \frac{1}{2}$.

(ii)

$$\begin{aligned} (X * Y) * Z &= \lambda\big(\lambda \mathbf{x} + (1-\lambda)\mathbf{y}\big) + (1-\lambda)\mathbf{z} \\ &= \lambda^2 \mathbf{x} + \lambda(1-\lambda)\mathbf{y} + (1-\lambda)\mathbf{z} \end{aligned}$$

$$\begin{aligned} X * (Y * Z) &= \lambda \mathbf{x} + (1-\lambda)\big(\lambda \mathbf{y} + (1-\lambda)\mathbf{z}\big) \\ &= \lambda \mathbf{x} + \lambda(1-\lambda)\mathbf{y} + (1-\lambda)^2 \mathbf{z} \end{aligned}$$

$$\begin{aligned} (X * Y) * Z - X * (Y * Z) &= (\lambda^2 - \lambda)\mathbf{x} + \big((1-\lambda) - (1-\lambda)^2\big)\mathbf{z} \\ &= \lambda(1-\lambda)(\mathbf{z} - \mathbf{x}) \end{aligned}$$

They are distinct when $\lambda \neq 0,\ \lambda \neq 1,\ X \neq Z$

(iii)

$$\begin{aligned} (X * Z) * (Y * Z) &= \lambda\big(\lambda \mathbf{x} + (1-\lambda)\mathbf{z}\big) + (1-\lambda)\big(\lambda \mathbf{y} + (1-\lambda)\mathbf{z}\big) \\ &= \lambda^2 \mathbf{x} + \lambda(1-\lambda)\mathbf{y} + (1-\lambda)\mathbf{z} \\ &= (X * Y) * Z \end{aligned}$$

$$\therefore (X * Y) * Z = (X * Z) * (Y * Z)$$

$$
\begin{aligned}
(X * Y) * (X * Z) &= \lambda\big(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}\big) + (1-\lambda)\big(\lambda\mathbf{x} + (1-\lambda)\mathbf{z}\big) \\
&= \lambda\mathbf{x} + \lambda(1-\lambda)\mathbf{y} + (1-\lambda)^2\mathbf{z} \\
&= X * (Y * Z)
\end{aligned}
$$

$$
\therefore X * (Y * Z) = (X * Y) * (X * Z)
$$

(iv) Let $\mathbf{p}_n$ be the position vector of P_n. $\mathbf{p}_1 = \lambda\mathbf{x} + (1-\lambda)\mathbf{y}$
Assume $\mathbf{p}_k = \lambda^k\mathbf{x} + (1-\lambda^k)\mathbf{y}$ is true

$$
\begin{aligned}
\mathbf{p}_{k+1} &= \lambda\mathbf{p}_k + (1-\lambda)\mathbf{y} \\
&= \lambda\big(\lambda^k\mathbf{x} + (1-\lambda^k)\mathbf{y}\big) + (1-\lambda)\mathbf{y} \\
&= \lambda^{k+1}\mathbf{x} + (1-\lambda^{k+1})\mathbf{y}
\end{aligned}
$$

By mathematical induction, $\mathbf{p}_n = \lambda^n\mathbf{x} + (1-\lambda^n)\mathbf{y}$ is true
When $0 < \lambda < 1$,

$$
XP_n : P_nY = (1-\lambda^n) : \lambda^n
$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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